



Guideline

Linezolid

Scope

Site	Service/Department/Unit	Disciplines
SCGH and OPH (i.e. SCGOPHCG)	Organisation Wide	Medical, Nursing, Pharmacy

Aim

Practical guideline aimed at clinicians to ensure the safe use of linezolid, including dosing and monitoring appropriate to the SCGOPHCG patient population.

Summary

- Linezolid is an antibiotic effective against a range of Gram-positive bacteria, Nocardia and Mycobacteria (1).
- It is important for the treatment of multi-resistant bacteria with limited options available e.g. VRE (vancomycin resistant Enterococcus) and MRSA (methicillin resistant *Staphylococcus aureus*).
- Although generally well tolerated linezolid use is associated with a range of serious adverse effects that require close monitoring and urgent management.
- While it should always be prescribed with the assistance and approval of Infectious Disease specialists, health care workers involved with the prescription and administration of linezolid should be informed of the potential adverse effects and monitoring requirements.
- Red antibiotic: an approval code from infectious diseases or clinical microbiologist is required prior to commencing linezolid.
- It is strongly recommended that patients planned for >2weeks duration be formally referred to Infectious diseases.

Action

- Linezolid is an oxazolidinone antibiotic that acts on bacterial ribosomes to disrupt protein synthesis (2). It is used for the treatment of VRE, MRSA, Nocardia and Mycobacterial infections (1).
- Linezolid is not metabolised by the cytochrome P450 pathway so there are no cytochrome-related drug interactions. However, other drug interactions do exist. Refer to 'Contraindications, precautions and drug Interactions' section for more detail.

Adverse Effects

- Adverse effects can occur at any time point, but the risk increases with duration of use and older age (3,4).

Table 1: Adverse Effects of Linezolid

Severity	Adverse Effects
Severe	Lactic Acidosis (risk factors: pre-existing renal insufficiency, hepatic impairment, low body weight, metformin use).
	Myelosuppression (thrombocytopaenia is the most likely manifestation which can usually be reversed with linezolid cessation or reduced dose).
	Serotonin Syndrome (increased risk with co-administration with drugs that increase serotonin levels. The risk is low, but a risk assessment needs to be made, see monitoring section below)
	Optic Nerve Damage (degradation of colour vision and visual acuity).
	Peripheral Neuropathy (can be irreversible and severe).
Common (>1%)	Nausea, vomiting and diarrhoea
	Abdominal pain
	Myelosuppression
	Headache
	Hepatic dysfunction
	Oral candidiasis
Less common (<1%)	Hypertension (avoid tyramine-rich foods)
	Rash, pruritis and eosinophilia
	Injection site reactions

	Peripheral neuropathy
	Hyponatraemia (including SIADH)
	Hypoglycaemia
	Insomnia
	Tongue discolouration and taste disturbance

Contraindications, precautions and drug interactions (3,4, 5,6)

Contraindications:

- Hypersensitivity or allergy to linezolid or any other component of the product
- Concomitant use of a monoamine oxidase inhibitor or within two weeks of taking a monoamine oxidase inhibitor, due to potential to cause serotonin toxicity. Examples include phenelzine, tranylcypromine, moclobemide, rasagiline, selegiline, safinamide.

Precautions:

Conditions

- Uncontrolled hypertension (or associated conditions such as hyperthyroidism, thyrotoxicosis or phaeochromocytoma).
- Myelosuppression (anaemia, neutropenia, thrombocytopenia)
- Phenylketonuria (in relation to the oral suspension) – the liquid formulation contains 20mg phenylalanine per 5mL (in the form of aspartame) which may be harmful to patients with phenylketonuria
- Carcinoid syndrome
- Diabetes mellitus: risk of hypoglycaemia induced by linezolid therapy. Additionally, the IV formulation contains 13.7g glucose per 300mL.

Concomitant therapies

- Treatment with myelosuppressive drugs or therapy can increase the risk of thrombocytopaenia and anaemia.
- Co-administration with drugs that increase serotonin levels (e.g. SSRIs, TCAs, triptans, lithium, see appendix 1 for examples) can increase the risk of a **serotonin syndrome** as linezolid acts as a weak reversible non-selective monoamine oxidase inhibitor (MAOI). However, it is important to note that the incidence of serotonin toxicity has been reported to be very low (7) (see Monitoring section below*).
- Clarithromycin – may increase linezolid concentrations and increase risk for toxicity

- Metaraminol / noradrenaline / adrenaline – as linezolid is a weak nonselective inhibitor of MAO it may reduce metabolism of noradrenaline to cause severe hypertension
- Pseudoephedrine / phenylephrine – as above, combination with linezolid can lead to cumulative effects of catecholamines with risk of hypertension
- Rifampicin – may decrease linezolid concentrations and overall clinical effect
- Rizatriptan, sumatriptan, zolmitriptan – their concentrations may be increased in the presence of linezolid with potential risk of ischaemia. Use naratriptan instead.

Pregnancy and breast feeding (8,9)

Limited information available: single case reports of healthy infants following second trimester maternal use and one case series of 5 (9). Has not caused congenital malformations in animals. Use alternative if possible.

Food Precautions (10)

- Tyramine-containing food or beverages – inhibition of tyramine metabolism by linezolid may result in potential increase in blood pressure

Examples of foods containing high tyramine content are in Appendix 1.

Dosing**Administration (5,6)**

Oral tablets:

- Can be administered with or without food
- May be crushed for administration via an enteral feeding tube (if the oral suspension is not available)

Oral suspension (granules) - 100mg/5mL

- Oral granules require reconstitution using 123mL of water (added in two separate, equal aliquots to produce a final volume of 150mL.
- Contents must be shaken thoroughly to produce a uniform suspension
- Prior to each use the bottle should be inverted several times to gently mix the contents to ensure an accurate dose is provided (do NOT shake)

Parenteral linezolid

- Intravenous solution (600mg/300mL premixed infusion bag)
- Suitable as an intravenous infusion, administered over a period of 30 to 120 minutes.

Not suitable for intramuscular, subcutaneous or intravenous bolus injection

Dosing Recommendations:

- The oral bioavailability of linezolid is close to 100% therefore the oral formulation is the preferred route of administration unless there is an issue regarding absorption or swallowing difficulties.
- Due to large interindividual serum level variability therapeutic drug monitoring (TDM) is recommended for all patients, but in particular for patients who have renal impairment, low or high body weight, are critically unwell or have an expected treatment duration longer than 14 days.

Table 2: Recommended Starting Dose for Linezolid

Indication:	Renal Function* eGFR $\geq$ 60mL/min	Renal Function* eGFR < 60mL/min (or weight <55kg)
Gram positive infections e.g. Enterococcus, Staphylococcus, Nocardia spp.	600mg 12 hourly	600mg 12 hourly on day 1 300mg 12 hourly thereafter
Mycobacterial Infection	600mg daily	300mg daily

Infectious diseases input should always be sought on dosing schedule and duration due to the risks of toxicity.

*Note: dose adjustment for renal impairment is debated. The product labelling states no dosage adjustment is required for renal impairment (based on an early PK study of n=24 patients given only a single 600mg linezolid dose), however there are now multiple studies showing renal insufficiency is strongly associated with linezolid overexposure and toxicity and a growing body of literature supporting dose adjustment to 300mg 12 hourly for eGFR <60 (11-15)). For example, a PK study by Crass et al (11) demonstrated that for patients with eGFR <60 mL/min, the risk of overexposure to linezolid at the standard dosage of 600 mg 12 hourly was >50%.

- Renal replacement therapy (16):
 - Intermittent haemodialysis - 100% 24 hourly (give the dose after dialysis)
 - Continuous renal replacement therapy (CRRT) – normal dosing as per $\geq$ 60mL/min

Note: both intermittent and continuous dialysis can lead to linezolid underexposure (below therapeutic levels), so in critically unwell patients higher doses should be considered in conjunction with TDM (17)

- Peritoneal dialysis - 100% 24 hourly. Dose reduction and TDM recommended due to increased risk of toxicity (18).

- Hepatic impairment

- Mild to moderate impairment (Child-Pugh A and B) – no dose adjustment
- Severe impairment (Child-Pugh C) – dose reduce to 300mg 12 hourly (dosing as per eGFR <60ml/min in Table 1)

Note: a study of patients with liver cirrhosis (mainly Child-Pugh B and C) showed a significantly increased incidence of overexposure (76.9% vs. 26.9%) in patients with cirrhosis compared to those without (19).

- Obesity

- There is conflicting data for those with high body weight (20)
- Levels in patients with raised BMIs appear to be similar to those of non-obese patients receiving standard dosing, so dose adjustments based on body weight alone are not routinely recommended
- If the patient is critically unwell and weight greater than 130kg a dose of 450mg 8 hourly should be considered, if practically feasible to give (20)

- Low body weight

- For patients with low body weight (defined as < 55kg) suggest reducing dose to 300mg 12 hourly (15)

- Critically unwell

- Higher doses should be considered in critically unwell patients.
- If renal function > 10ml/min, or receiving CRRT support then consider increasing the dose to 450mg every 8 hours, if practically feasible (15, 20)
- Monitor full blood count closely and perform linezolid TDM

- ECMO: case reports only, reporting both under and over exposure on standard dosing (17)

Monitoring (3, 4, 5, 13, 15)

Baseline:

1. Check if the patient is on a serotonergic drug but given the very low risk of serotonin syndrome a risk assessment should be performed using the information in the note below (see note below)**.
2. Check for other drug interactions
3. Give patient dietary advice (for inpatients consider dietician involvement)
4. Baseline FBC, U&Es, lactate, LFTs
5. Routine ophthalmologic and neurological assessment for patients expected to receive greater than 28 days of linezolid.
6. Ongoing monitoring as per Table 3. Suggest print Table 3 and place in patient's medical record.

**Note: The incidence of serotonin toxicity has been reported in a systematic review (7) to be 0.0050% with linezolid monotherapy and 0.0134% when linezolid combined with a serotonergic drug. All cases which discontinued linezolid and serotonergic agent at signs and symptoms of toxicity found symptom resolution, usually within 24-48hours. It is therefore no longer routinely required for patients to cease serotonergic medications for short durations of linezolid therapy where they can be carefully monitored and report symptoms (<2 weeks), but more detailed patient education and monitoring should occur for patients who take linezolid combined with a serotonergic agent. Cessation may still be required, for example in patients on multiple serotonergic agents or if unable to comply with more detailed education and monitoring requirements. If cessation of serotonergic agents occurs, a 2-week washout is preferable.

Table 3. Linezolid Monitoring

Adverse Effect	Monitoring	Comment
Serotonin syndrome	Monitor for tremulousness, agitation, hyperreflexia, tachycardia or diaphoresis.	Advise patient on symptoms to report immediately.
Pancytopenia	Perform a baseline FBC and then weekly to monitor for myelosuppression	Carefully monitor for a drop in platelet count (even within the normal range) as an early indicator of supratherapeutic levels. Patients with other risk factors for myelosuppression should be monitored more closely.
Renal and hepatic function	U&E: individual risk assessment (affects dose given) LFTs: rare cause of LFT abnormalities	Perform LFTs if develop nausea or upper abdominal pain
Lactic Acidosis	Perform baseline lactate and monitor serum lactate twice weekly in patients with pre-existing renal insufficiency, hepatic impairment, low body weight, metformin use or other risk factors for linezolid toxicity.	All patients: if nausea, reduced serum bicarbonate or unexplained acidosis occurs organise urgent lactate level. Can occur early in the treatment course.
Neuropathy	Monitor for signs and symptoms of peripheral neuropathy.	Advise patient on symptoms to report.
Optic neuropathy	Monitor visual acuity, visual field and colour vision for use greater than 28 days.	Baseline then monthly ophthalmology assessment, Advise patient on symptoms to report.
Therapeutic Drug Monitoring	TDM should occur for all patients receiving more than 2 weeks.	See TDM section.

Therapeutic Drug Monitoring (13, 15)

1. Consider TDM in all patients.

Monitoring of linezolid should be considered in all patients due to the large serum level interindividual variability and evidence base demonstrating improved clinical outcomes in those with TDM (15). In particular TDM should be used if:

- Prolonged duration of therapy (beyond 10 to 14 days)
- Physiological extremes e.g. morbid obesity or low body weight, older age >65years old (21)
- Renal impairment (<60ml/min)
- Renal replacement therapy (at risk of insufficient linezolid exposure if on intermittent or continuous haemodialysis, at risk of overexposure if peritoneal dialysis)
- Augmented renal function (at risk of insufficient linezolid exposure)
- Hepatic impairment (particularly cirrhosis Child-Pugh C)
- Critically unwell
- Receiving higher or lower than standard doses of linezolid
- Taking medications known to interact with linezolid

2. Collect 12-hour trough level at least 48h after commencement of linezolid.

The sample collected should be a 12-hour trough level (i.e. just prior to next dose for BD dosing and at 12 hours for once daily dosing***), collected **at least 48hrs** (preferably 3-5days) following either initiation of therapy or any dose adjustment. Note that steady state may be reached after this time i.e. levels might increase after this.

The trough concentration is chosen because this level has been shown to correlate with drug exposure (area under the curve) and therefore clinical drug efficacy. Declining platelet count (even within normal range) is a marker of toxicity and should prompt review of dosing and a linezolid trough level.

***Expert opinion suggests using a 12hour level rather than a trough for once daily dosing. Experience suggests using a 24hour trough level in this situation leads to low levels and increasing the dose frequently causes toxicity.

3. Consider weekly levels during treatment courses >2weeks duration.

The blood sample should be collected into either a red (serum and clot activator) or dark green (lithium heparin, no gel) collection tube.

RED
Serum + clot activator



OR

DKGNLITH

Lithium heparin (no gel)



Table 4. Target trough concentration ranges for linezolid

INDICATION	TARGET TROUGH (12-hour) CONCENTRATION
Gram-positive bacterial infections	2 to 6.5mg/L
Mycobacterial infections	2mg/L
Interpretation of the level should be performed collaboratively between the ID pharmacist, Clinical Microbiologist, and ID physician	

CAUTION - Levels exceeding 8mg/L are considered in the toxic range and associated with greater risk for thrombocytopenia.

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Authorisation & Version Control

Policy Sponsor:	SCGOPHCG Director of Clinical Services			
Policy Contact:	SCGOPHCG Medicines and Therapeutics Committee Executive Officer (Senior Pharmacist)			
Date First Issued:	13/10/2025	Last Reviewed:	13/10/2025	Review Date: 13/10/2028
Version No.	1	Reference #:	MMG059	
Approved by:	SCGOPHCG Antimicrobial Stewardship Committee			Date: 20/08/2025
Endorsed by:	SCGOPHCG Medicines and Therapeutics Committee			Date: 13/10/2025
Acknowledgements:	Authors: Jason Seet (Infectious Diseases Pharmacist) Dr Julie Hart (Microbiology Consultant)			
Standards Applicable	NSQHS Standards 			

Appendix 1

Drugs that can increase serotonin levels.

Examples:

- SSRIs (e.g. citalopram, dapoxetine, escitalopram, fluoxetine, fluvoxamine, paroxetine, sertraline)
- TCAs (e.g. amitriptyline, clomipramine, dosulepin, doxepin, imipramine, nortriptyline)
- SNRIs (e.g. desvenlafaxine, duloxetine, venlafaxine)
- Agomelatine, mianserin, mirtazapine, reboxetine, vortioxetine
- Others: St John's Wort, dextromethorphan, fentanyl, pethidine, tramadol, lithium, methylene blue, phentermine, amphetamines

Appendix 2

Tyramine containing foods and beverages:

- Aged, dried or pickled meats (e.g. mortadella, salami, pepperoni, meat jerky)
- Pickled or smoked fish
- Fermented tofu products
- Mature cheeses such as cheddar, camembert, gouda, gruyere, parmesan, provolone, stilton, aged fetta, goats cheese
- Buttermilk
- Probiotics (e.g. Yakult)
- Fermented or pickled vegetables (e.g. sauerkraut, Kim chi)
- Overripe fruit
- Fish sauce, shrimp paste, protein supplements
- Avoid large amounts of yeast extract spreads (e.g. Vegemite, Promite, Marmite), soy sauce, teriyaki sauce, hoi sin sauce
- Miso soup
- Artisan breads (e.g. sourdough bread)